

Homework Classical Physics, Fall 2023

Prof. J. Fritz, Research III Room 56, jfritz@constructor.university.de

Problem Set 02.1 (L03 & L04)

Due on Thursday, 21. September 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - motion (4 points)

A car is sitting at a stop light and then accelerates from rest when the light turns green. The acceleration of the car is a combination of forward acceleration but also slowing down due to drag forces. The final total acceleration has the form $a(t) = b/(t + T)^3$, with constants $b = 2000 \text{ m}\cdot\text{s}$ and $T = 10 \text{ s}$.

- a) What are the cars velocity and position as a function of time?
- b) What are the cars velocity and position at $t = 5.0 \text{ s}$?